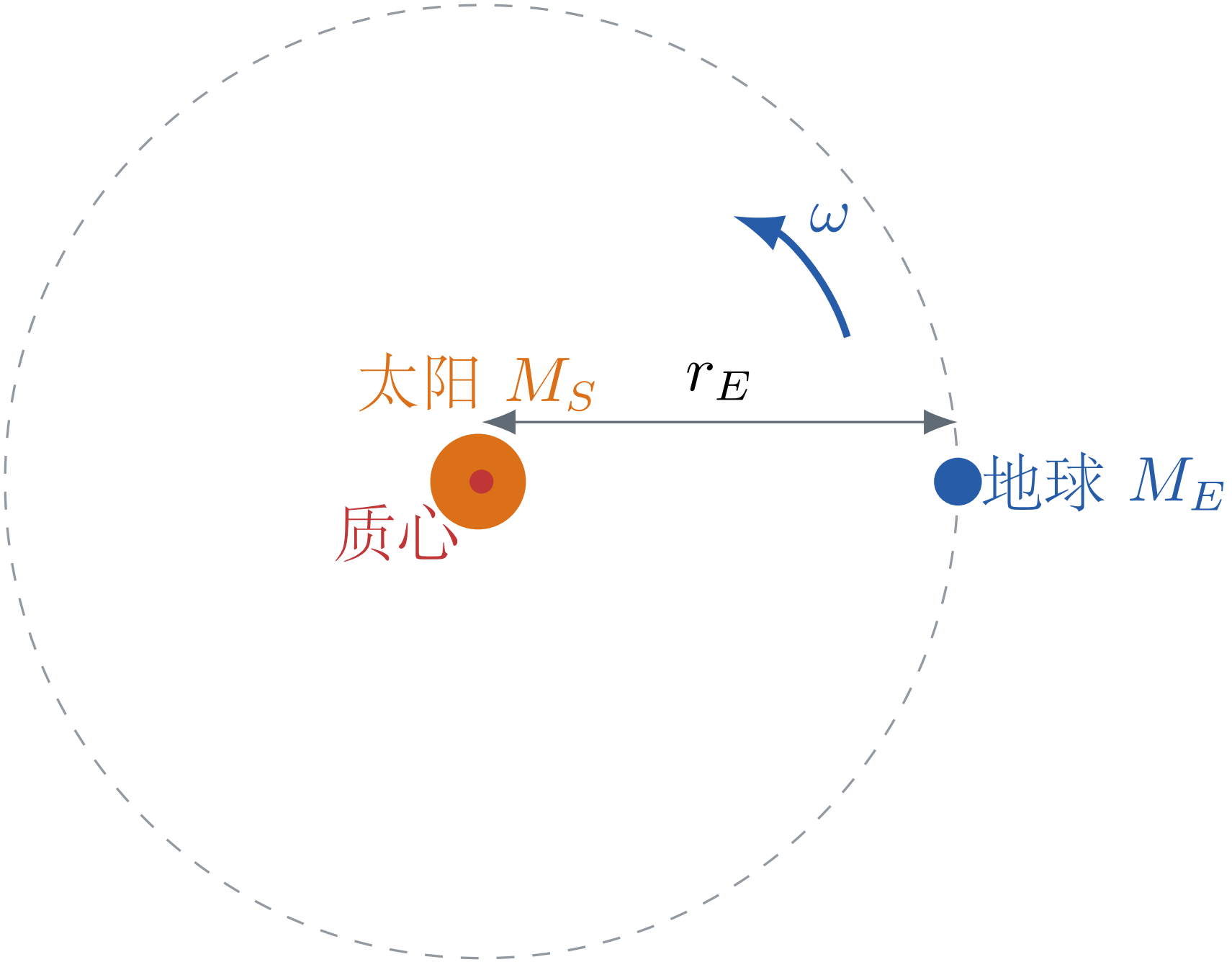
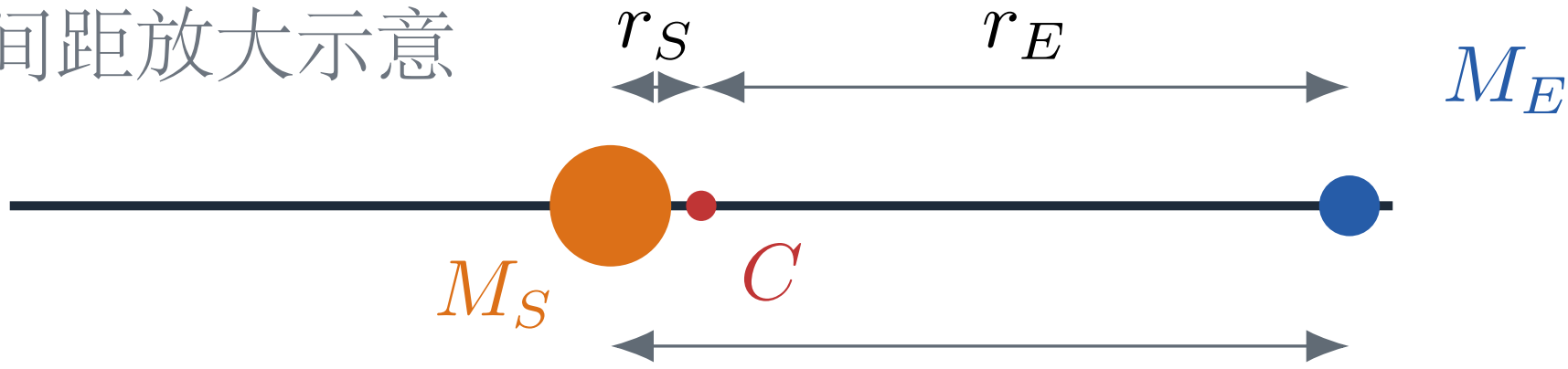


恒星光谱的周期性红移与蓝移

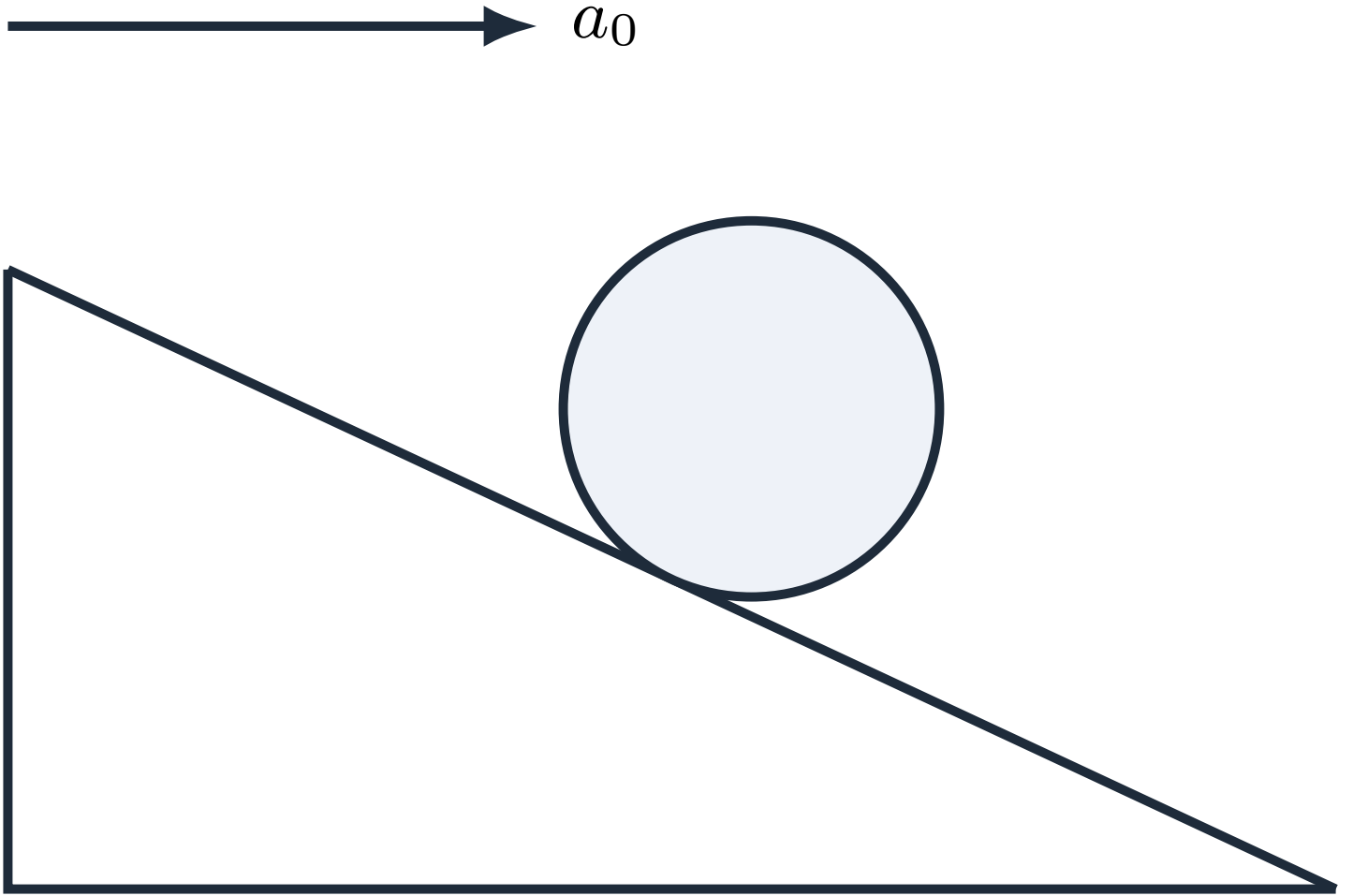
对应太阳的往返径向速度

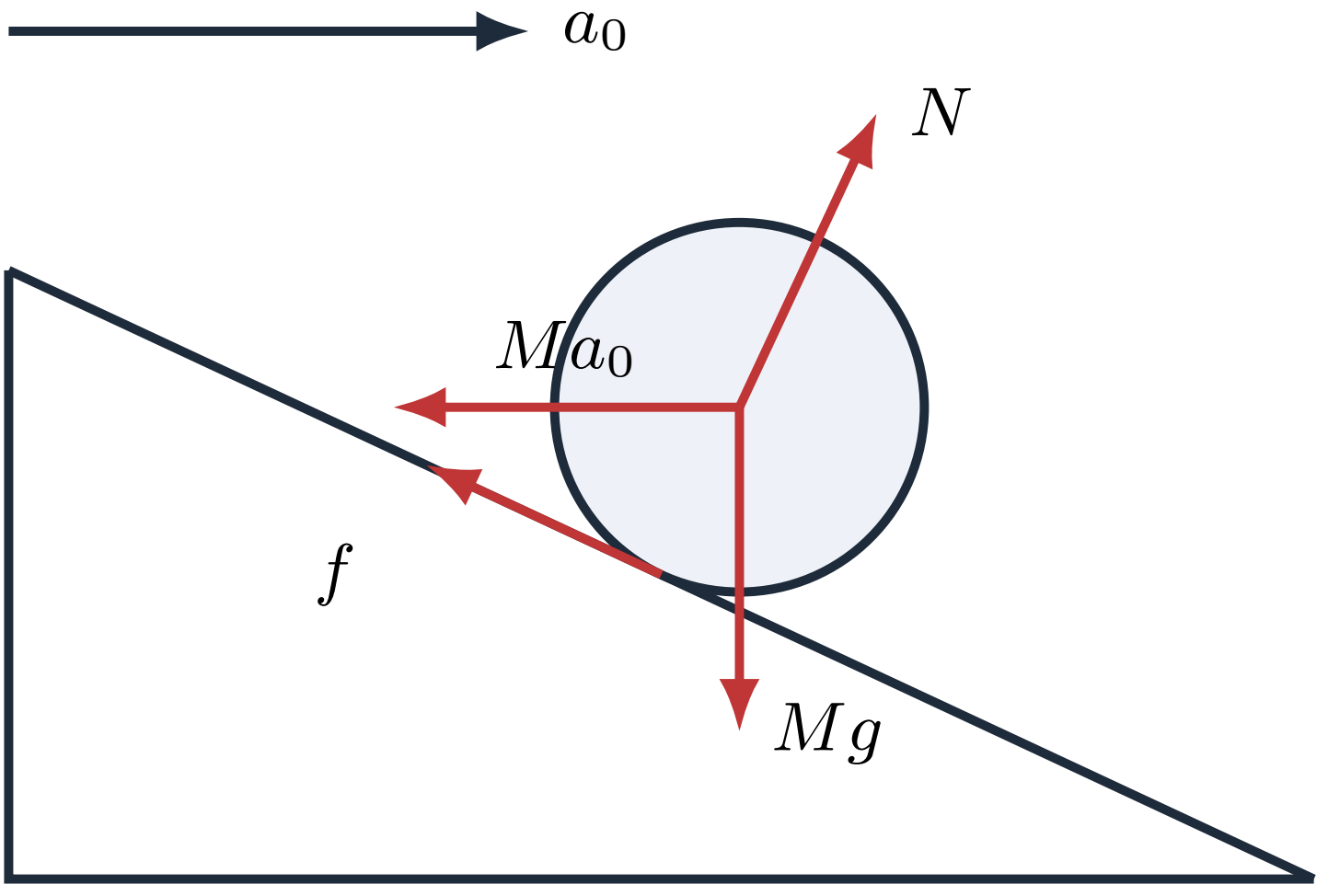
间距放大示意



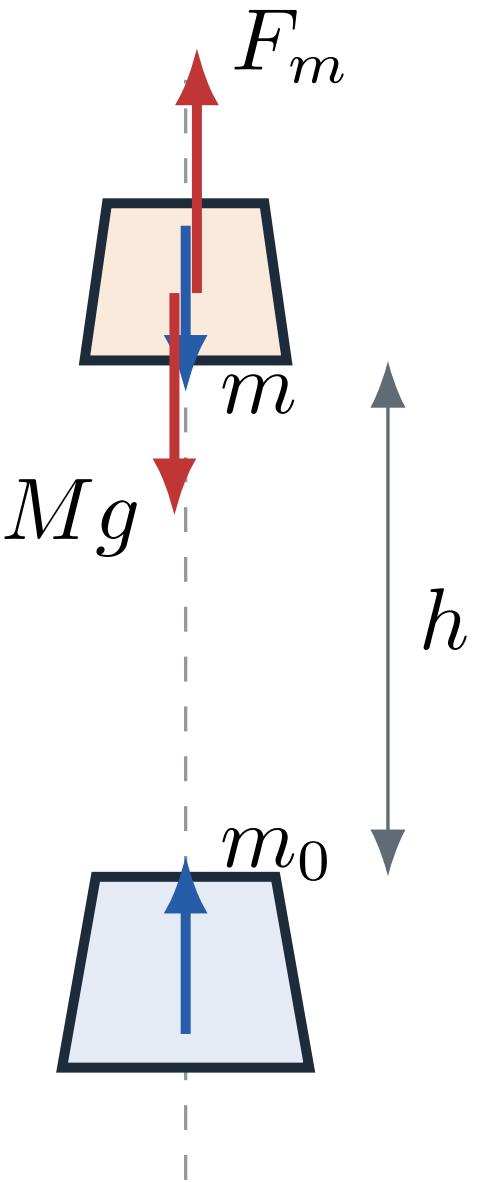
$$M_S r_S = M_E r_E \quad r_S + r_E = L$$

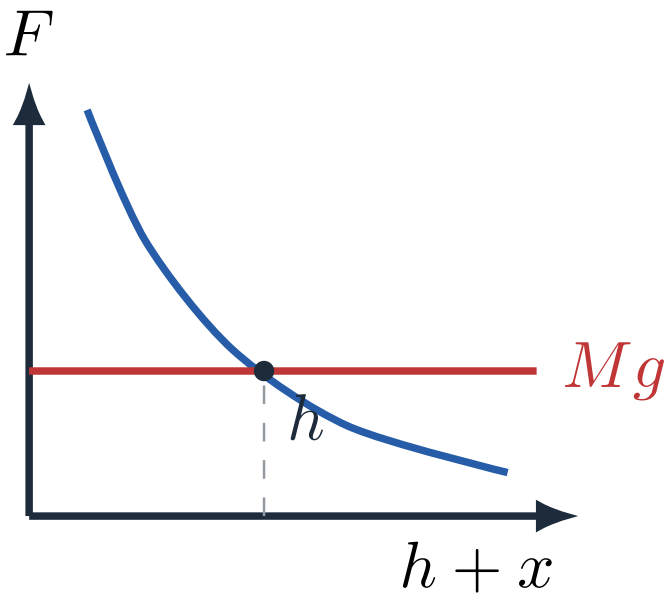
$$v_S = \omega L \frac{M_E}{M_S + M_E}$$


a_0

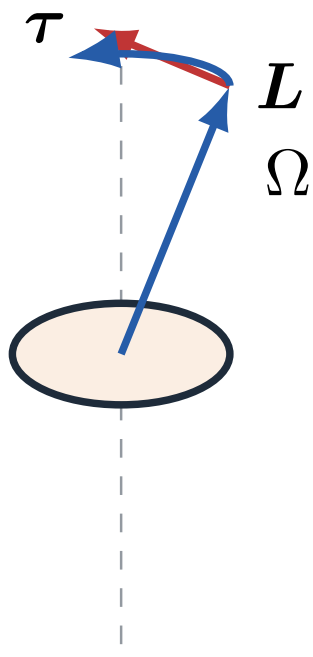


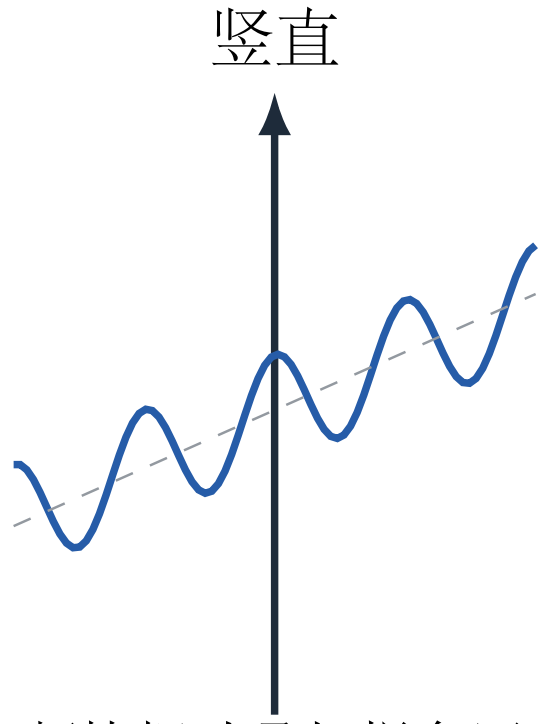
$$a_r = \tfrac{2}{3}(g\sin\theta - a_0\cos\theta)$$


F_m
m
Mg
h
m_0



$$F_m \propto (h+x)^{-4}$$

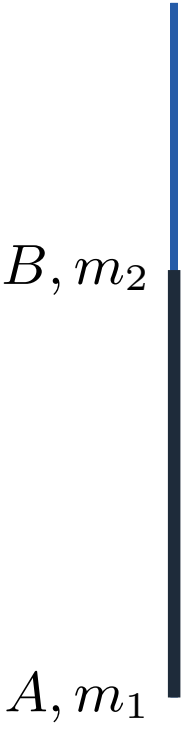




高频快振动叠加慢角运动

$$B_c = \omega\sqrt{2I_0B_0/m};\ \ B_1 > B_c$$

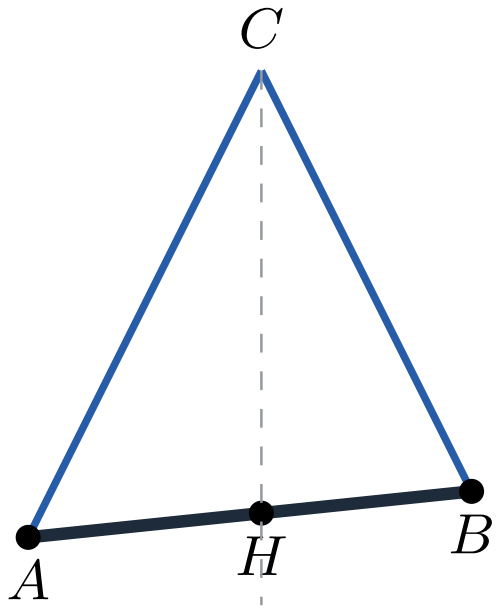
$$\Omega_{\pm} = \frac{\omega_0 \pm \sqrt{\omega_0^2 - 4mB_0/I_0}}{2}$$



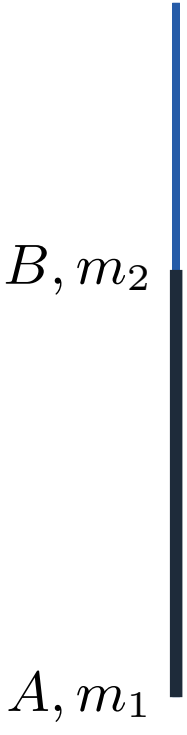
I：$\delta < c/a$ 稳定



II：稳定



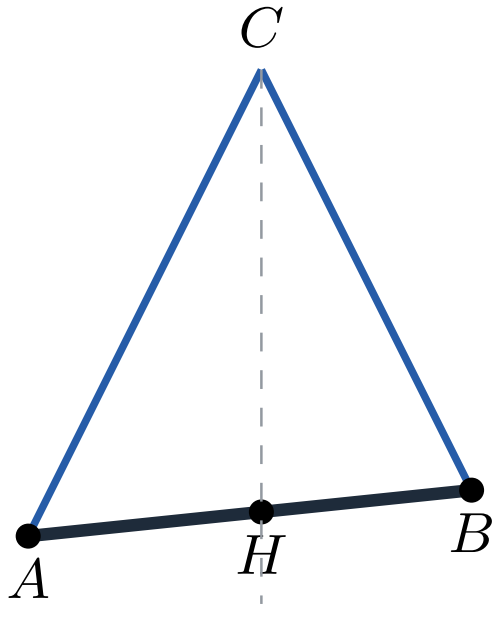
III：鞍点



I：$\delta < c/a$ 稳定

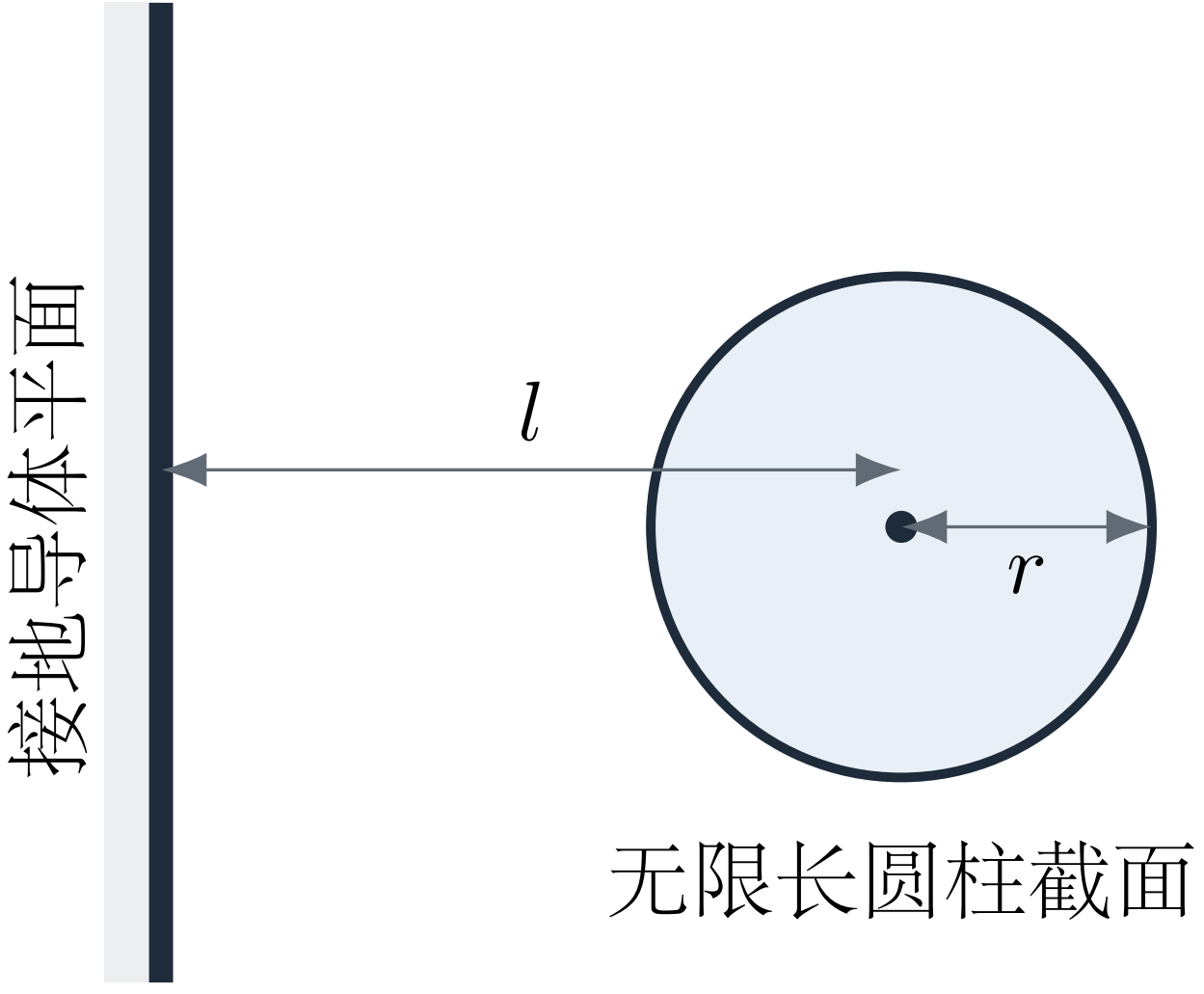


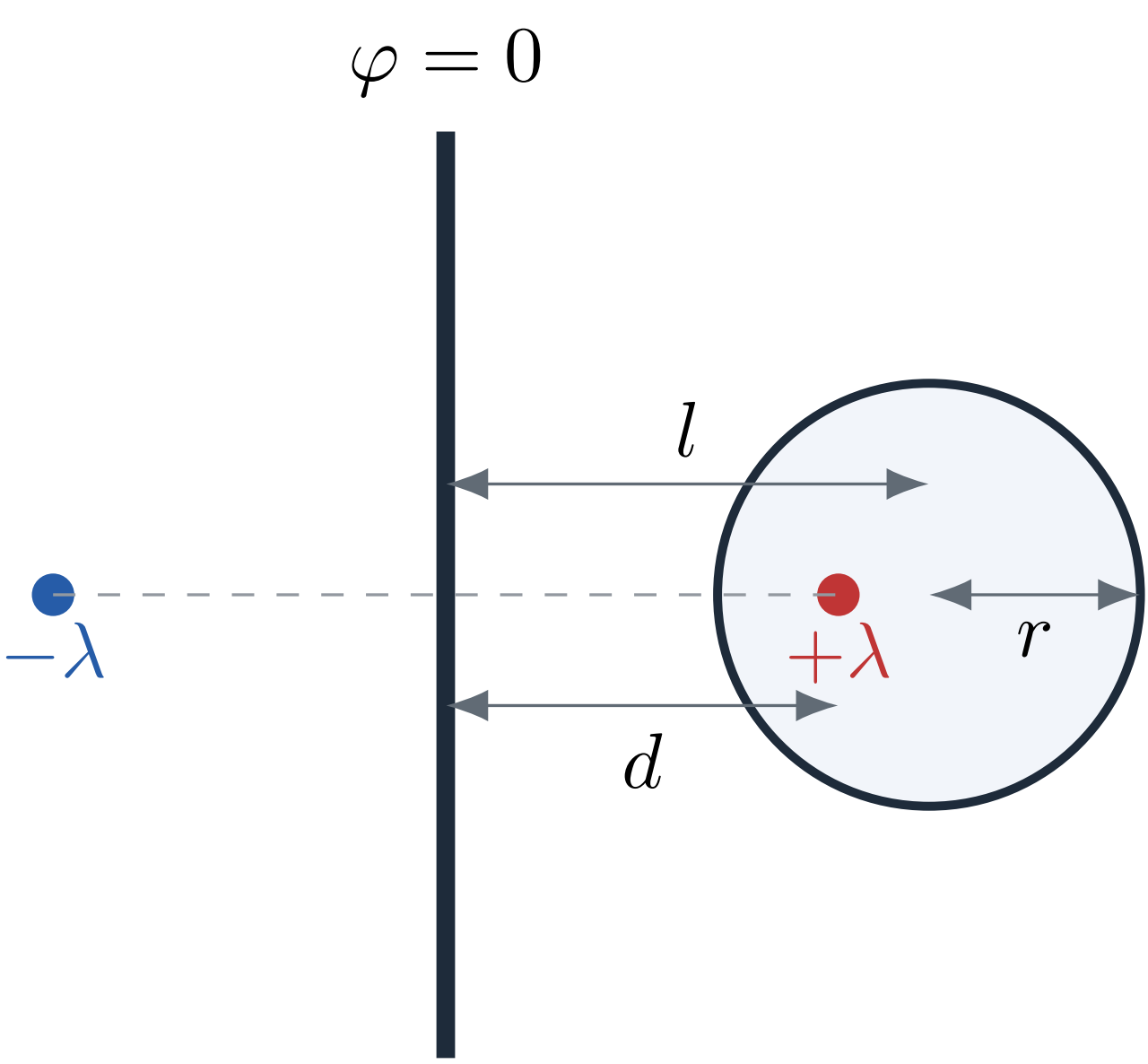
II：稳定



III：鞍点

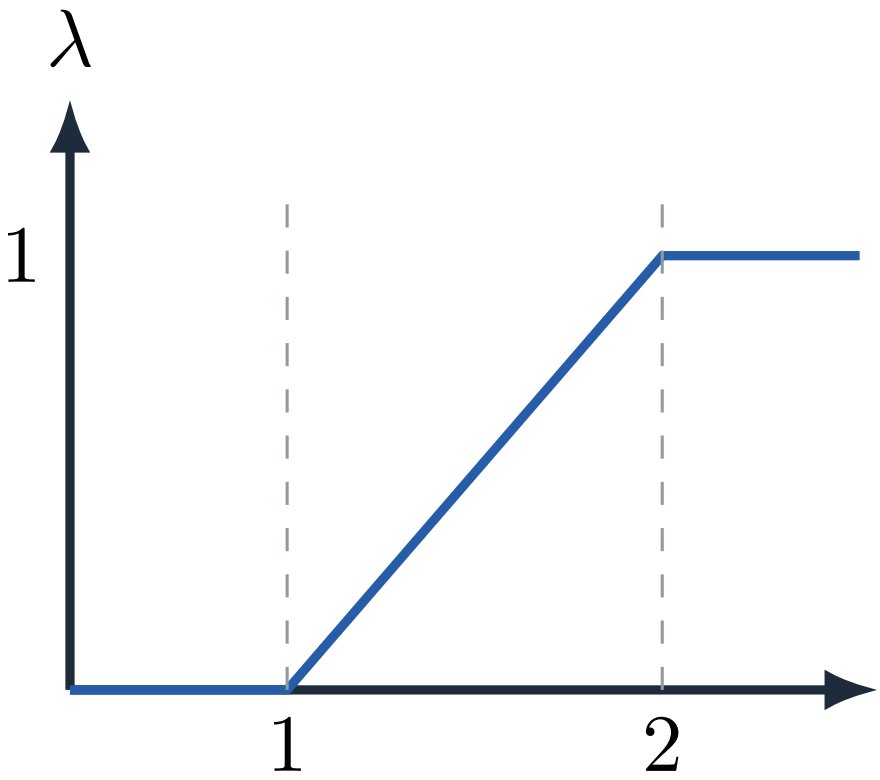
$\det K_I \propto M - a(m_2 - m_1)/c$；二自由度判定稳定性


接地导体平面
l
r
无限长圆柱截面

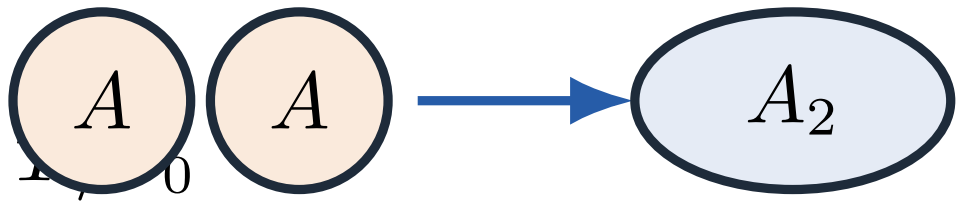


$$d^2 = l^2 - r^2$$

$$C' = \frac{2\pi\varepsilon}{\operatorname{arccosh}(l/r)}$$

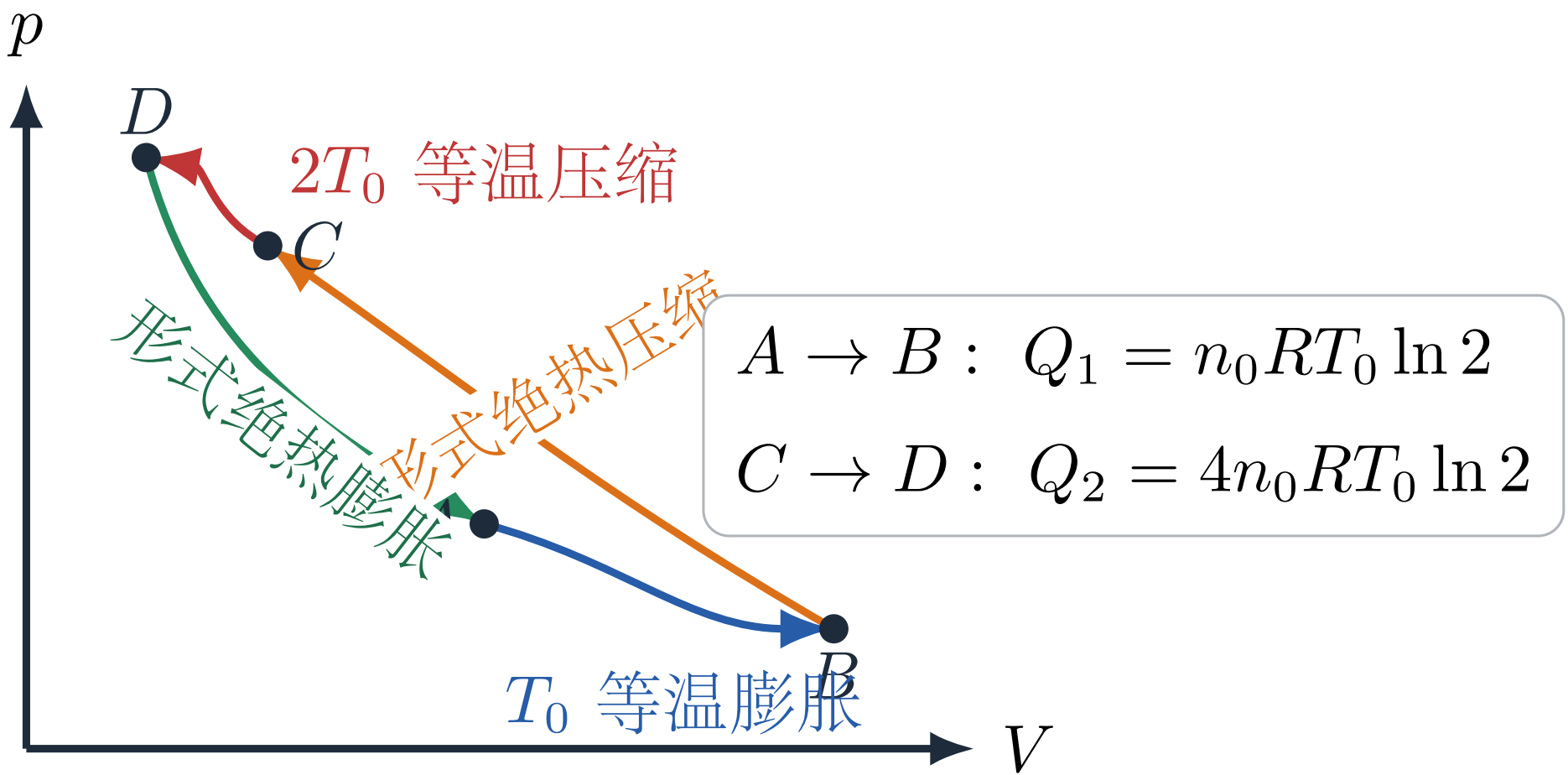


$$2A \rightleftharpoons A_2$$



$$n = (1+\lambda)n_0 = n_0 T/T_0$$

$$pV = n_0 R T^2/T_0$$



$$A \to B:\ Q_1 = n_0 R T_0 \ln 2$$

$$C \to D:\ Q_2 = 4 n_0 R T_0 \ln 2$$

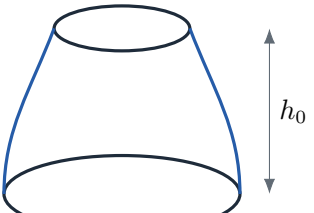
体积轴为示意比例



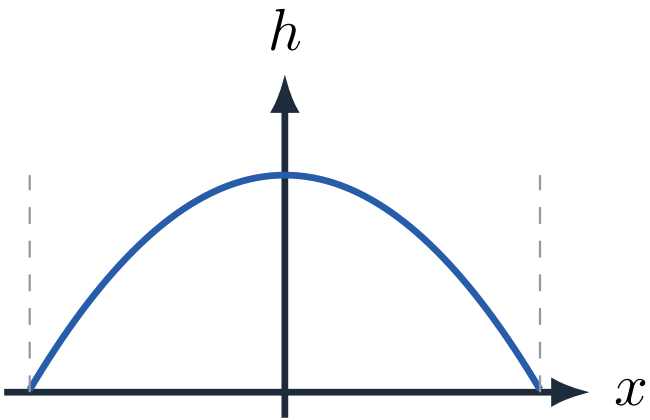
横波膜元



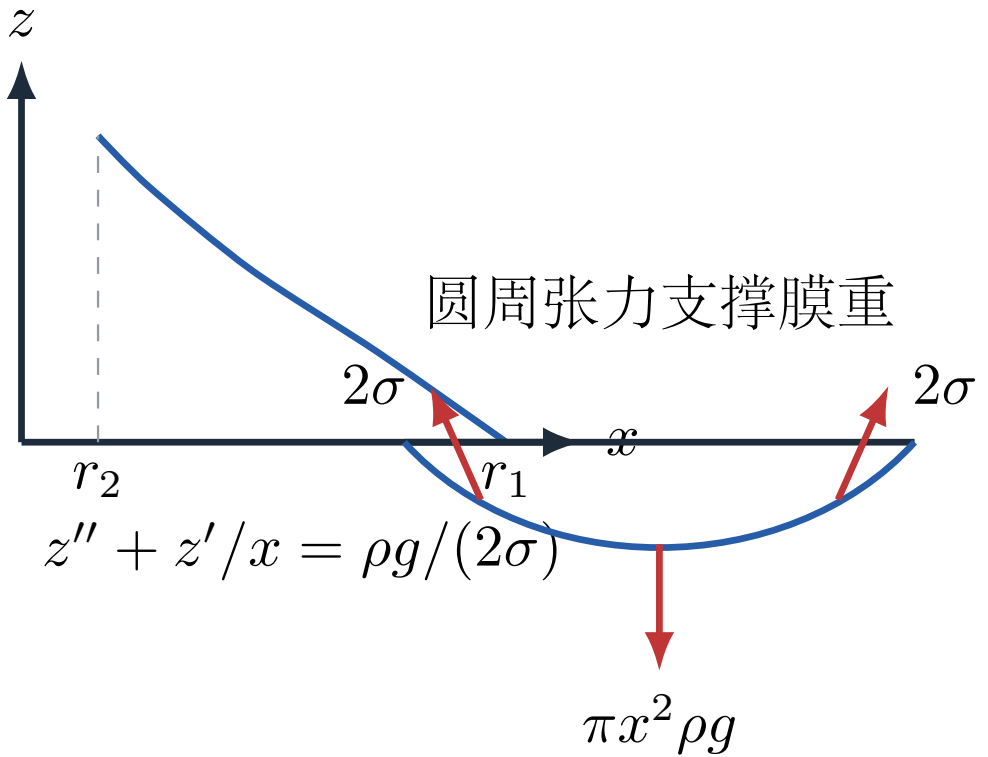
水平圆膜重力下沉

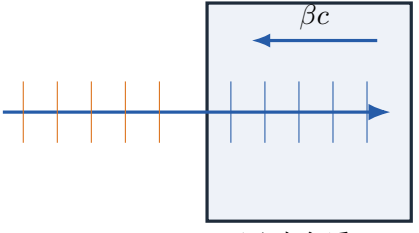


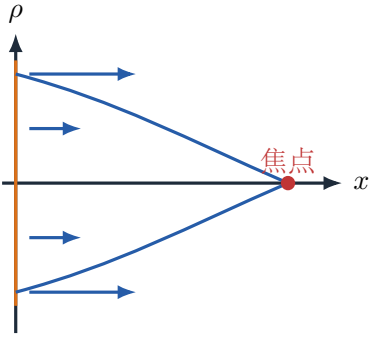
同轴双圆环



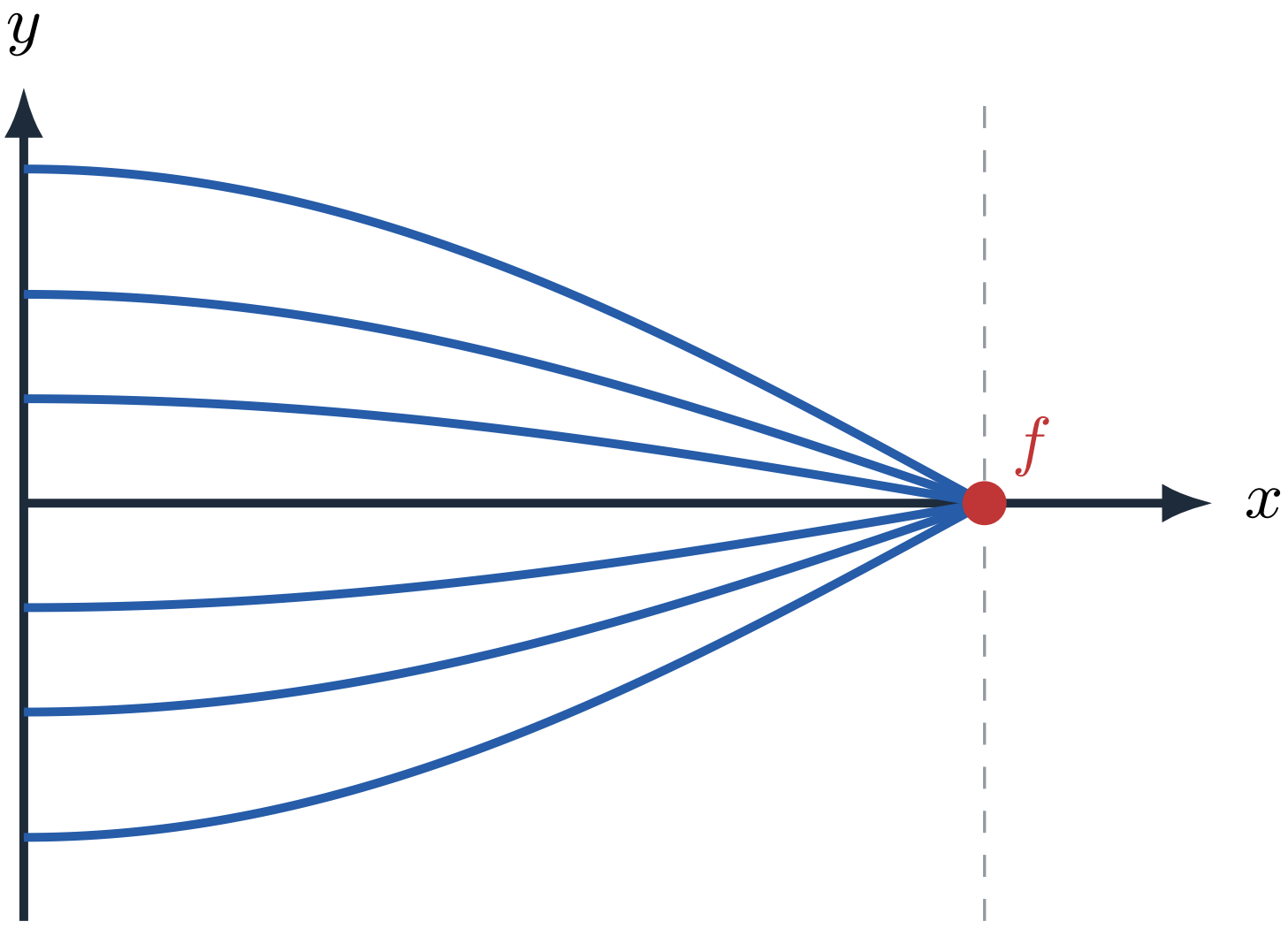
$$h(x) = \frac{\rho g}{8\sigma}(r^2 - x^2)$$







$$u(\rho) = c\rho^2/\rho_0^2$$



$$y'' + \frac{2(n^2-1)}{n\rho_0^2} y = 0$$

$$f/\rho_0 = \frac{\pi}{2}\sqrt{\frac{n}{2(n^2-1)}}$$